PROJECT 2

# Customer Segmentation and Retention Analysis

GitHub Project README

| INTERNSHIP TRACK | DATASET |
| --- | --- |
| Data Analytics | Online Retail II - UCI Machine Learning Repository |
| **TOOLS** | **CORE METHODS** |
| Python, Pandas, Matplotlib, Seaborn, Google Colab / Jupyter | RFM scoring, customer segmentation, cohort retention |

PROJECT SNAPSHOT

**779,425 valid sales records | 5,878 customers | 5 quality checks passed**

Prepared: 7 November 2026

Team roles are described as Member 1 and Member 2. Add the actual names and repository URL before final submission if your college requires them.

# 1. Project Overview

This project analyses retail transactions to understand customer purchasing behaviour and customer retention. It uses RFM analysis (Recency, Frequency and Monetary value) to group customers and cohort analysis to study how customer activity changes after a customer's first purchase.

## Project objectives

• Clean and prepare transaction data for analysis.

• Calculate transaction revenue and customer-level RFM measures.

• Assign customer scores and create customer segments.

• Build monthly cohort customer-count and retention-percentage matrices.

• Visualise cohort retention with a heatmap and recommend retention actions.

# 2. Dataset

The source workbook contains two sheets: Year 2009-2010 and Year 2010-2011. The combined source data contains 1,067,371 rows and 8 columns.

| Field | Description |
|---|---|
| Invoice | Invoice or transaction identifier |
| StockCode | Product code |
| Description | Product description |
| Quantity | Number of items in the transaction line |
| InvoiceDate | Transaction date and time |
| Price | Unit price |
| Customer ID | Customer identifier |
| Country | Customer country |

Source: Chen, D. (2012). Online Retail II [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C5CG6D

# 3. Data Cleaning Summary

| Cleaning item | Result |
|---|---|
| Original records | 1,067,371 |
| Duplicate rows removed | 34,335 |
| Missing Customer IDs removed | 235,151 |
| Cancelled invoice records removed | 18,390 |
| Invalid quantity / price records removed | 70 |
| Final valid transaction rows | 779,425 |
| Total transaction revenue | 17,374,804.27 |

Missing product descriptions were replaced with 'Unknown'. Cancelled invoices were identified by invoice values beginning with C. Valid sales were restricted to positive quantities and positive prices. Revenue was calculated as Quantity multiplied by Price.

# 4. Analysis Methods and Results

## RFM analysis

Recency is the number of days since the customer's latest purchase, measured relative to the day after the last transaction in the dataset. Frequency is the number of distinct invoices. Monetary is the customer's total transaction revenue. Scores use quartiles on a 1-4 scale: lower recency receives a higher score, while higher frequency and monetary values receive higher scores.

A project-specific, rule-based segmentation assigns customers to Loyal Customers, Big Spenders, At Risk Customers, Recent Customers or Others. These labels are analytical groupings, not externally validated classifications.

## RFM segment summary

| Segment | Customers | Avg. Recency | Avg. Frequency | Avg. Monetary |
|---|---|---|---|---|
| At Risk Customers | 1,470 | 514.03 | 2.02 | 712.90 |
| Big Spenders | 204 | 42.17 | 2.33 | 2,416.85 |
| Loyal Customers | 2,051 | 29.62 | 12.94 | 6,485.55 |
| Others | 1,469 | 225.47 | 4.01 | 1,541.84 |
| Recent Customers | 684 | 39.81 | 1.61 | 390.32 |

## Cohort analysis

Each customer's cohort month is the month of their first recorded purchase. The cohort index measures the months between that first purchase month and each later transaction month. The matrix counts unique customers in each cohort-month cell. Retention percentage is the cell count divided by the cohort's month-zero count, multiplied by 100.

# 5. Retention Recommendations

• Loyal Customers: recognise repeat purchases with loyalty benefits and relevant exclusive offers.

• At-Risk Customers: test personalised win-back messages and targeted offers; prioritise high-value customers within this group.

• Big Spenders: offer relevant premium products or personalised service; high spending alone does not mean loyalty.

• Recent Customers: provide helpful onboarding and product suggestions that encourage a second purchase.

• All segments: monitor repeat-purchase and cohort-retention trends to check whether campaigns work.

# 6. GitHub Repository Guide

## Suggested repository structure

```
Project_2_Customer_Segmentation/
├── data/
│   └── cleaned_online_retail.csv.gz
├── notebooks/
│   └── Project_2_Customer_Segmentation.ipynb
├── outputs/
│   ├── customer_rfm_analysis.csv
│   ├── customer_segment_summary.csv
│   ├── cohort_customer_counts.csv
│   ├── cohort_retention_percentage.csv
│   └── cohort_retention_heatmap.png
├── reports/
│   ├── data_cleaning_log.md
│   ├── data_quality_checks.py
│   ├── customer_retention_plan.md
│   └── project_conclusion.md
├── README.md
└── README_Project_2_Customer_Segmentation.pdf
```

The original Excel workbook is approximately 43.5 MB and should not be uploaded through GitHub's standard browser uploader. Link to the UCI source dataset instead. The compressed cleaned CSV was previously measured at approximately 12.45 MB. If the team chooses to store three compressed parts, list all three part names in the README and explain how to combine them in order.

## How to run

Download the UCI dataset or use the team's cleaned CSV. Open the notebook in Google Colab or Jupyter, install the packages below if needed, update the dataset path, and run the cells in order. Review the data-quality checks, RFM summary and cohort heatmap.

## Install dependencies

```
pip install pandas openpyxl matplotlib seaborn
```

# 7. Team Responsibilities

| Member | Main responsibilities |
|---|---|
| Member 1 | Dataset loading and cleaning, data-quality checks, revenue calculation, RFM analysis and customer segmentation. |
| Member 2 | Cohort month and index, cohort customer-count matrix, retention percentage matrix, heatmap and customer-retention plan. |
| Both members | Notebook integration, result review, documentation and final submission. |

Replace the role labels with the members' actual names if required. Record each person's commits under their own GitHub account.

# 8. Limitations

The transactions cover 2009-2011 and may not represent current retail behaviour. Records without Customer IDs are excluded from customer-level calculations. RFM groups use simple quartile scores and project-specific rules. Cohort retention shows observed repeat purchasing but does not explain why customers return or leave. Proposed retention actions should be evaluated with future data or controlled experiments.

# 9. Conclusion

The project prepared 779,425 valid sales records and analysed 5,878 identified customers. RFM analysis produced five customer groups, while cohort analysis provided monthly customer-count and retention views. Together, these methods describe purchasing patterns and support targeted customer-retention planning.